SI 618 Winter 2025 Project Description & Part I Instructions

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Introduction

The team project component of SI 618 is designed to encourage you to both apply and extend the skills, tools, techniques, and knowledge that you've gained in this course to a new problem.

Overall, the goal of this project is to provide you with an opportunity to bring your unique interests, creativity and ingenuity into alignment with one or two team members and demonstrate your team's ability to apply data science skills and concepts on public datasets of your choice.

In particular, the project should show off your team's ability to take **a minimum of two data sets** possessing different features and clean and manipulate them in order to extract a useful byproduct: something more than you could have gotten from either data set by itself. The cleaning and manipulation could involve filtering, format conversion, handling missing or noisy data, matching records from one data source with corresponding records in the other, and so on, and must make significant use of the programming tools and techniques covered in this course. You will use this combined data set for subsequent components of the project assignment (i.e. the data analysis and machine learning components).

The project consists of **4** components:

(1) data description and basic manipulation,
(2) data analysis,
(3) application of machine learning techniques and
(4) presentation

Thus, the project serves as a comprehensive assessment of your work in the course.

You are not expected to use advanced tools and techniques from courses that are not covered, and in fact you are discouraged from doing so.

Part I: Team Formation, Data Description and Basic Manipulation

Team Formation

Team formation and the selection of project topics and datasets are student responsibilities. We suggest that you advertise your interests via Slack.

We expect that **all teams will consist of two to three (2-3) students**. The maximum team size is three (3) students. Working alone or working in groups of more than three is ***not*** permitted. It has been our experience that working in pairs or groups of three balances high-quality work with the amount of work each person needs to do to complete the project.

Data Selection

The single best predictor of performance on the project component of this course is the selection of (at least two) datasets in which you and your teammates are genuinely interested.

You are welcome to search for any datasets that interest you, but the following sources are good starting points as they contain a large number of diverse datasets.

Kaggle: https://www.kaggle.com/datasets

awesome-pubic-datasets repository (github) (many more data sets, including: social networks, sports, museums, machine learning)

KDNuggets: https://www.kdnuggets.com/datasets/index.html

World Development Indicators:
https://datatopics.worldbank.org/world-development-indicators/

Health Data : https://healthdata.gov

CDC dataset: https://data.cdc.gov

data.gov (over 200,000 data sets on topics including: public safety, health, education, climate)

FiveThirtyEight (a news site that makes article data on politics, sports, science/health, economics, and culture available to the public)

Once you have selected your datasets (or perhaps before), **generate 3 "real-world" questions about the data**. Your questions should be sufficiently broad that they cannot be trivially answered (e.g. "how many bikes were shared in New York in 2002?").

Data Manipulation

Once you have selected your datasets, and thought about some questions you might want to answer with your data, you should perform some basic data manipulation to merge them, as well as start thinking about how you'll handle problematic aspects of the data, such as missing values, outliers, etc. When we assess this component, we will be looking for evidence of the following:

1. The data has been successfully imported and displayed in the notebook.
2. Datasets have been successfully merged
3. The data is in a 'tidy' and usable format. In other words, you have created one dataset that can be used for analysis. This might involve cleaning the data, dealing with missing values, creating new columns based on existing data, etc.
4. The ability to perform some basic data manipulation techniques. This might include creating new columns, filtering out rows, or other techniques that you have learned in class.
5. The ability to use the tools we have learned in class to visualize your data in some appropriate way. This might include using libraries like matplotlib or seaborn to create plots, or using pandas to group data in a variety of ways.

6. You should demonstrate that you are able to produce a summary of the data, in a format that is easy to read and understand.

Note that this component of the project should stop short of conducting actual statistical analyses (i.e., only use techniques covered in the first four weeks of the course).

Submission Requirements

Your submission for this component will consist of a Jupyter notebook consisting of markdown and code cells that describe what you've done (markdown), how you've done it (code), and interpretation of the results (markdown).

Please use the following headings to structure your notebook:

Project Title
Provide a descriptive working title for your project.
Team Members
List each team member and include their uniqname
Overview
Give a high level description of your project
Motivation
Explain why you chose this particular topic for your project
Include the three "real-world" questions that you generated about the data, and be sure to explain what you hope to learn by answering them
Data Sources
List the two (or more) sources of data that you'll be using
Provide URLs where appropriate
Explain how the two (or more) datasets complement each other
Data Description
List the variables of interest, the size of the data sets, missing values, etc.
Data Manipulation
Mostly code in this section. This is where you merge your data sets, as well as create new columns (if appropriate)
Data Visualization

Produce 3 or more plots that help describe your data. Choose appropriate visualizations

Rubric

Grades will be allocated as follows:
Project Title: 2.5% (5 points)
Team Members: 2.5% (5 points)
Overview: 5% (10 points)
Motivation: 10% (20 points)
Data Sources: 10% (20 points)
Data Description: 20% (40 points)
Data Manipulation: 25% (50 points)
Data Visualization: 25% (50 points)
Each component will be graded on the following 5-point scale:

5: Excellent work, goes **beyond** stated requirements, shows innovation
4: Good work, meets stated requirements, does not offer particularly innovative approaches or scenarios
3: Fair work, meets some stated requirements, some errors and omissions
1-2: Inadequate work, meets only a few requirements, errors present
0: Work is absent

Changelog

2024.10.5.3.CT: Add questions to "Motivation" section